

ECE4132 Project 1 Report

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Note: Task 3 and 4 reports have been combined into this document

Task 3

(i) Optimal β ($\beta = 1$)

To obtain our optimal value of β , first consider the time-domain equation for the controller output:

$$\pi[k] = \beta\pi[k-1] + \kappa(e[k] - \alpha e[k-1])$$

Define a new parameter $\beta' = \beta - 1$. Then the equation can be rewritten as

$$\pi[k] - \pi[k-1] = \Delta\pi(k) = \beta'\pi[k-1] + \kappa(e[k] - \alpha e[k-1])$$

which can be seen as a difference equation approximation of some continuous-time differential equation with timestep $\Delta k = 1$.

Under this interpretation, the “rate of change” of $\pi(k)$ at timestep k depends on the price at timestep $k-1$, as well as an error term. Adjusting β therefore controls the steady-state error of the overall system response. If $\beta > 1$, then $\beta' > 0$, and the price tends to grow exponentially over time:

$$\frac{\Delta\pi}{\Delta k} \approx \frac{d\pi}{dk} \approx k\pi(k) + \text{error}(k)$$

for some constant $k > 0$ and some error function $\text{error}(t)$.

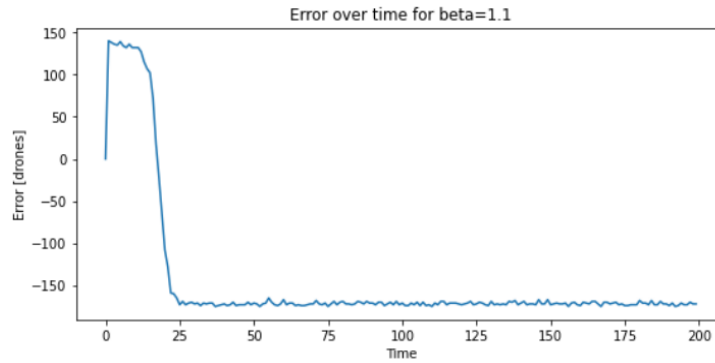


Figure 1: Error over time for $\beta > 0$, showing poor steady-state error

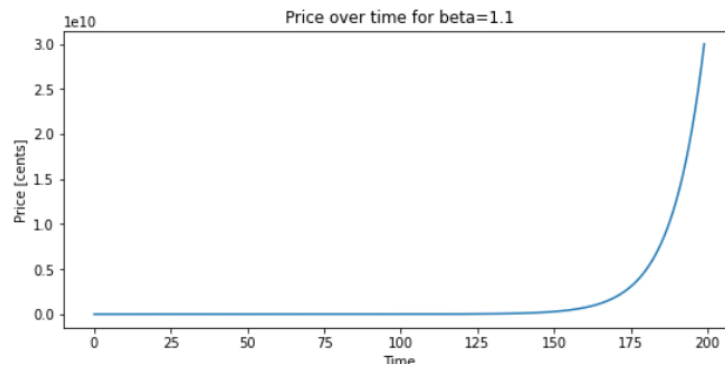


Figure 2: Price over time for $\beta > 0$, showing exponential growth

Due to the particular scaling of the logistic function used to model success probabilities, the error will only become negative for large prices (relative to the error); the error being bounded by $n_{orders} - n_{drones}$ (the number of orders less the number of drones) from below also means that the exponential behaviour will generally dominate for this choice.

Conversely, choosing $\beta < 1$ means $\beta' < 0$, so the controller tends to reduce the price; the steady-state error is therefore a balance between this downwards pressure on the price and the effect of the error driving the price upwards (with a low price, few drone owners will accept, so the error will generally be positive). The steady-state error, as before, will not be zero.

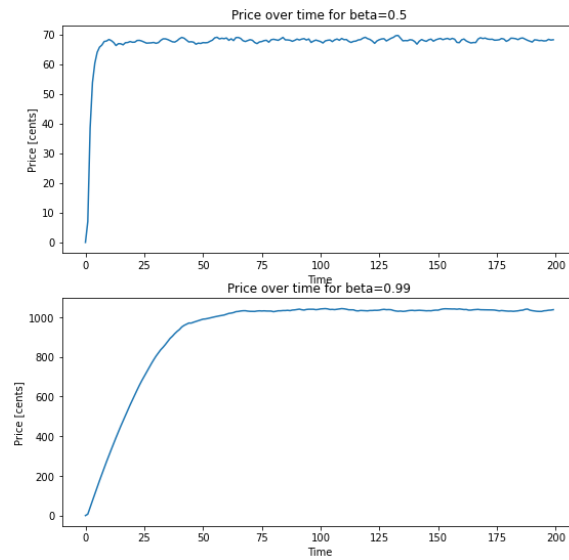


Figure 3: Price over time for various $\beta < 0$. Increasing β allows a higher steady-state price.

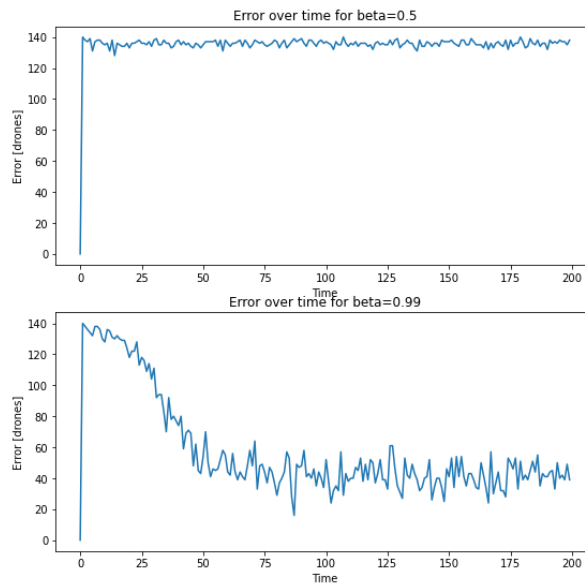


Figure 4: Error over time for various $\beta < 0$. Increasing β means less steady-state error.

Clearly, the optimal value for β appears to be $\beta = 1$, resulting in $\beta' = 0$. Intuitively, this means that the difference equation only depends on the error term:

$$\frac{\Delta\pi}{\Delta k} \approx \frac{d\pi}{dk} \approx \text{error}(k)$$

Consequently, the system will only reach steady state when the error term is driven to zero (which corresponds to past and current error summing to zero). Plots are shown below:

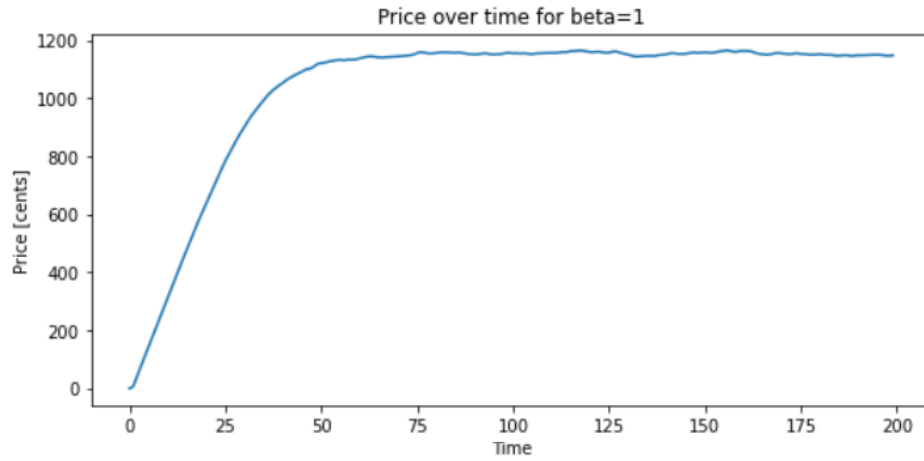


Figure 1: Price over time for $\beta=1$. Steady-state price now corresponds to minimal error.

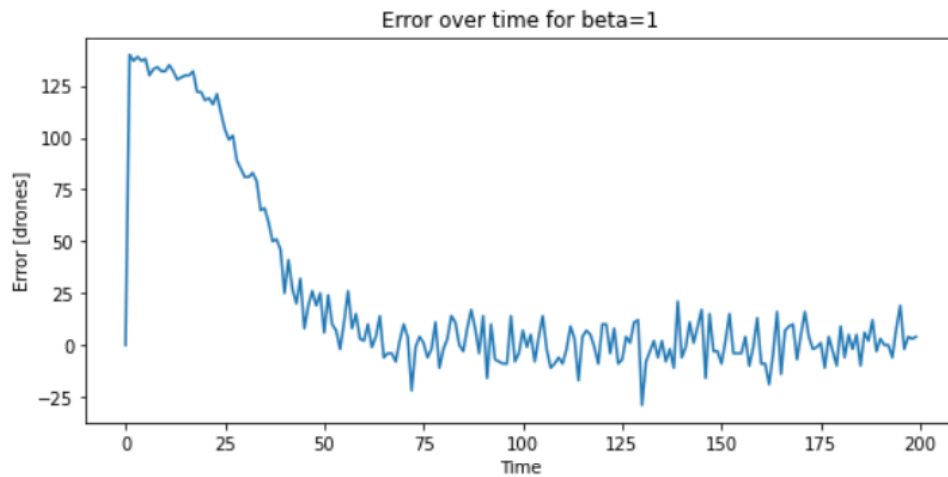


Figure 6: Error over time for $\beta=1$. Steady-state error (with noise) is now centred on 0.

Since the objective is to minimise the steady-state error, $\beta = 1$ was determined to be optimal.

(ii) Optimal κ ($\kappa = 0.05$)

Setting $\beta = 1$, we now consider the difference equation with the exact error term:

$$\pi[k] - \pi[k - 1] = \Delta\pi(k) = \kappa(e[k] - \alpha e[k - 1])$$

From this equation, we can intuitively see that κ controls the “sensitivity” of the system to the error signal by adjusting the scale of the price change in response to error. Larger κ amplifies the effect of error on price changes, and vice versa. (This also means that κ controls the rise time of the system, which is not too important for an offline system, but should be reasonably short.)

With this in mind, and taking into account the fact that the randomness inherent in the system generates high-frequency, low-amplitude noise in the error, we might expect that a relatively small, nonzero κ would result in a relatively low, stable error. Plots are shown below:

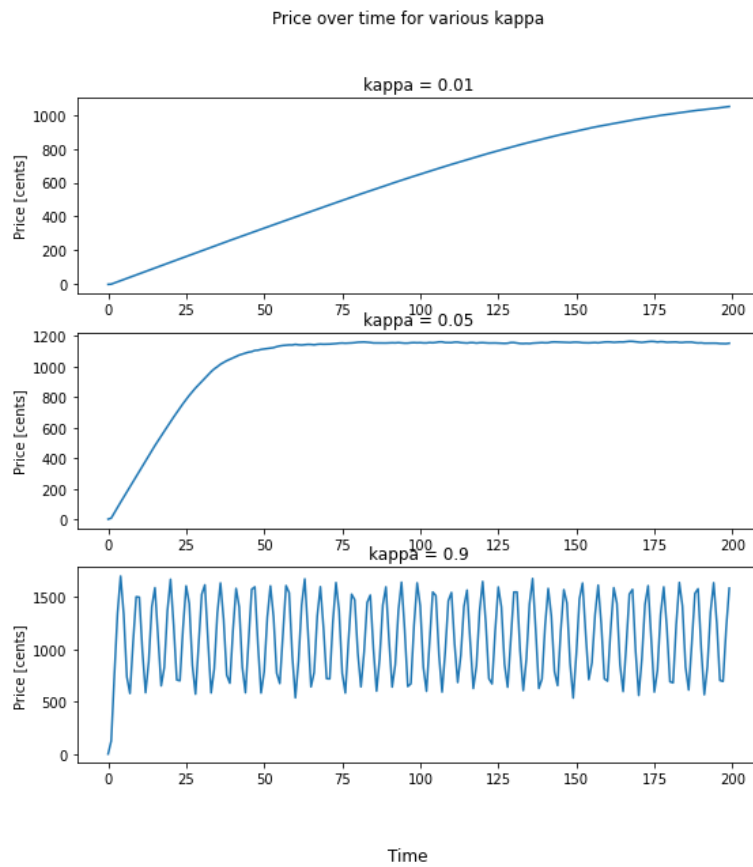


Figure 7: Price over time for various κ . Higher κ results in faster rise time and greater sensitivity to noise.

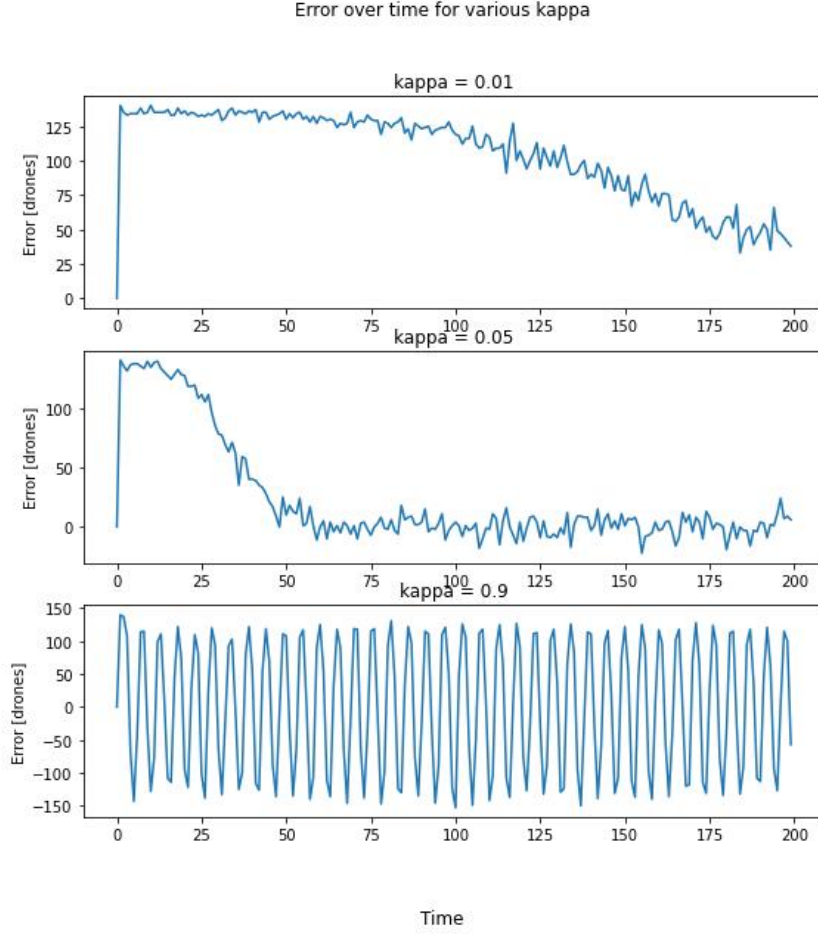


Figure 8: Error over time for various κ . Larger κ amplifies noise in error, but decreasing κ has diminishing returns.

$\kappa = 0.05$ was chosen as the optimal value through trial and error, based mainly on the observations that the resulting error signal does not oscillate significantly, and that smaller values of κ produce the same price estimate but took longer to converge.

(iii) Amount to offer to drone owners

The optimal amount to offer to drone owners was found to be \$11.60. This value was obtained by simulating the drone system once with the parameters determined above ($\alpha = -4.01, \beta = 1, \kappa = 0.05$), and taking the mean of the price outputs corresponding to the last 50 timesteps (timesteps 150 to 200). (The system was assumed to be in steady state after 100 timesteps, determined by visual inspection of the plots.)

(iv) Noise in $e[k]$ and $\pi[k]$

Regardless of the choice of parameters, not all error can be eliminated from the signals e and π . This is because the inherent randomness in the system – the number of drones dispatched Y , is a binomial random variable $Y \sim \text{Binomial}(n_{\text{drones}}, p)$, where the success probability parameter p is a function of the price π , the output of the provided logistic function. As a result, while

keeping the price fixed will result in a fixed value of p , this does not guarantee that the number of drone owners who accept will be the same each time. It should be noted, however, that while the noise in the error signal cannot be suppressed, the price signal can be smoothed by setting a suitable value for κ , as mentioned above.

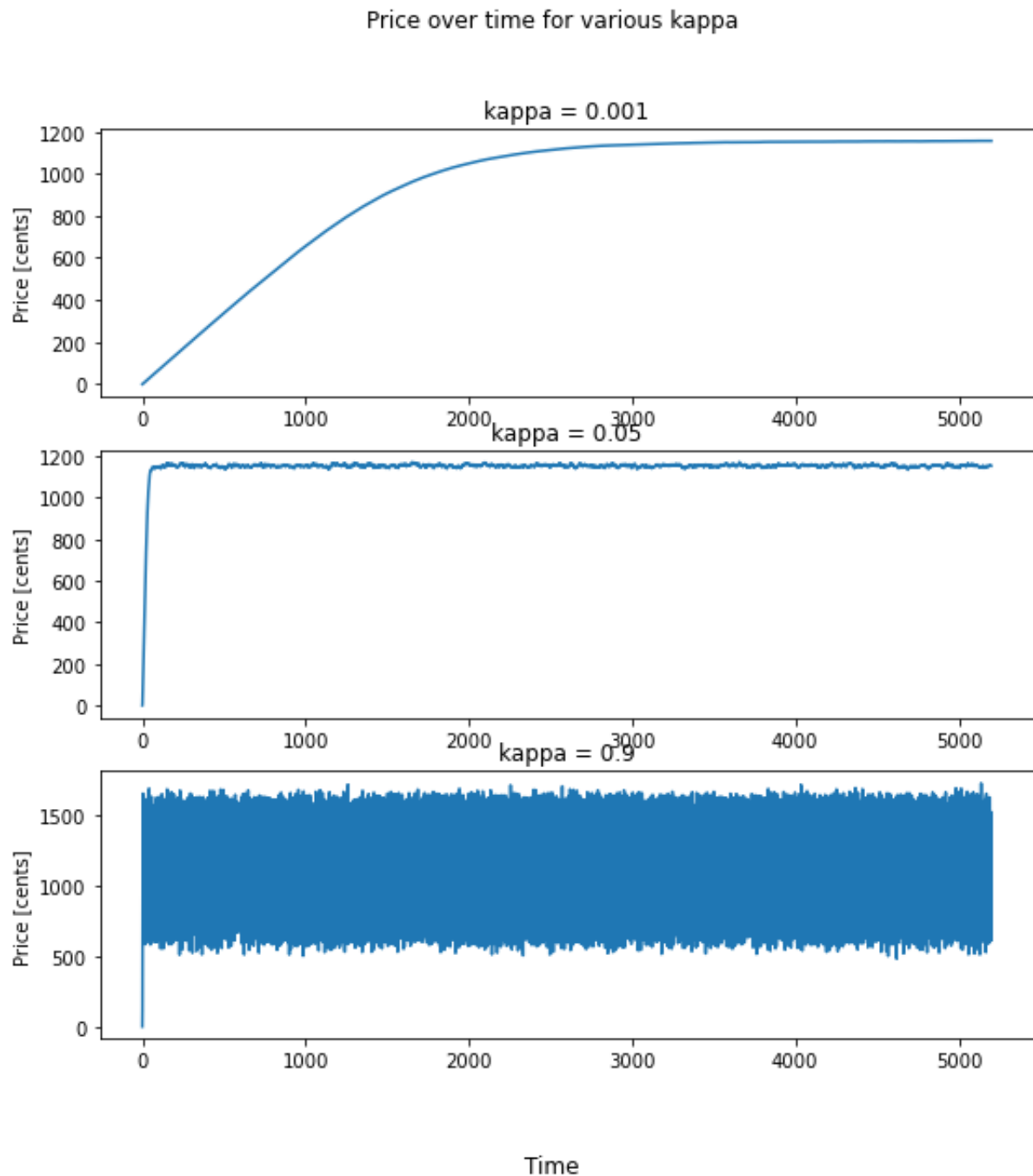


Figure 9: Price over time for various (different) κ . Choosing smaller κ clearly filters out a significant amount of noise.

Task 4

Predictability in terms of i) Mean of e and π and standard deviation from these means and ii) Revision of the price

We iterate over the system 300 times and average the signals for error, price and number of yes drones. As seen in figure 10 below, the noise from the error signal is significantly reduced but not fully eliminated due to the nature of random variables.

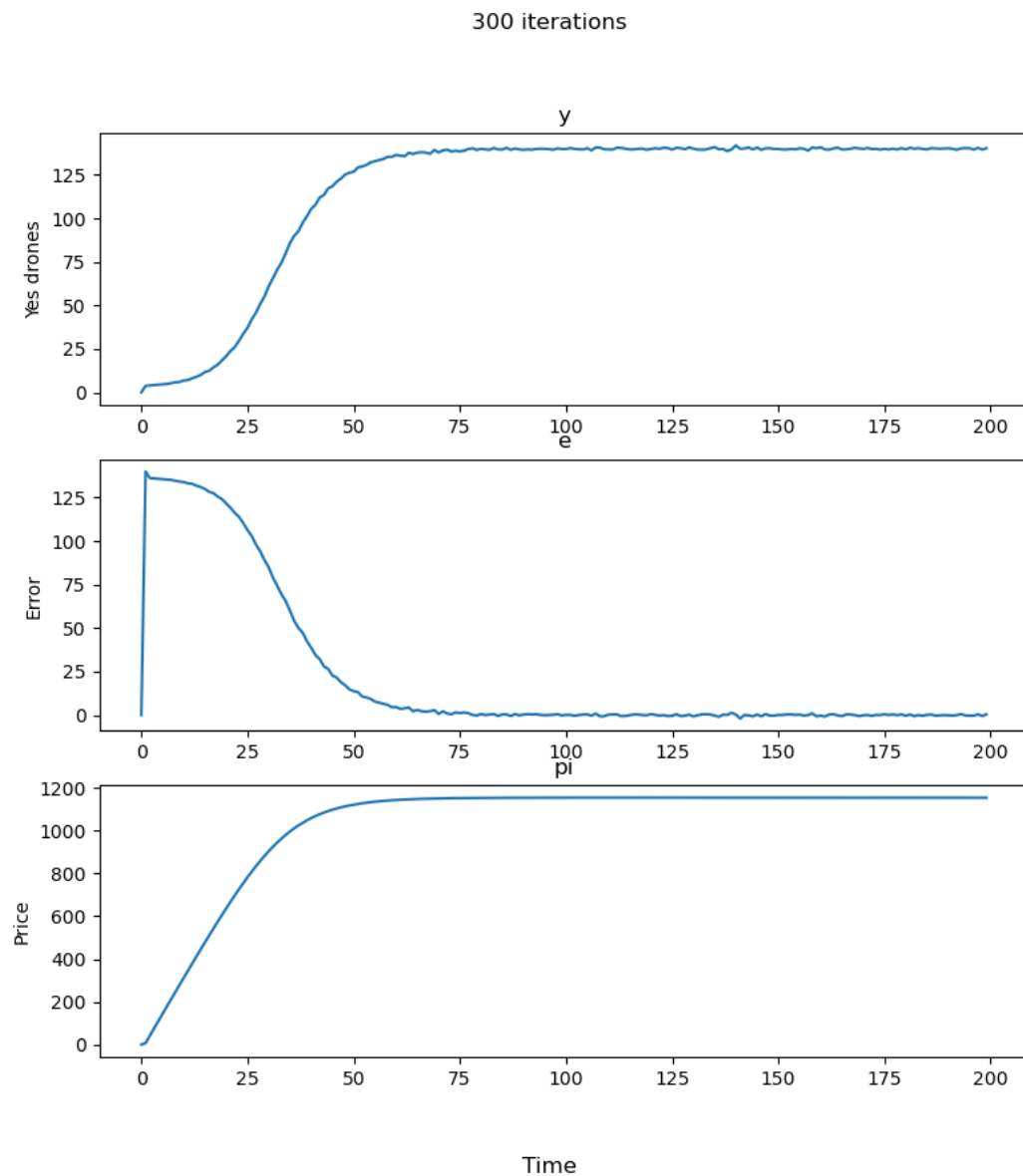


Figure 10 2: Mean values for price, error and number of yes drones after the system was run 300 times

Similar to task 3, the steady state is assumed to have been reached after 100 timesteps. When taking the mean of the last 50 timesteps, the optimal price was found to be \$11.55 with a standard deviation between iterations of 0.03. Likewise, the error after 300 iterations was found to have a mean of -0.02 with standard deviation 0.62. The

standard deviation of the error signal is greatly reduced compared to that of a single trial, which had a standard deviation of 8.76. While this does not tell us anything about a particular trial, it does tell us that on average, our specific controller outputs a price (\$11.55) such that the resulting error is centred around zero, with positive and negative deviations occurring with equal frequency and amplitude.

We can also see that, with the given parameters, the system's price output is extremely consistent despite the noise in the error signal – the price only deviates by 3 cents on average across 300 iterations of the simulation, each time converging on a price of ~\$11.55. Interestingly, this price is very close equivalent to the price for which the expected number of dispatched drones is 140.

Figure 11 below confirms the stability of the system. Given an initial price of \$50 which is well above the average of ~\$11.55 found above, the system converges to the steady state price. Note that the rate of convergence depends on κ .

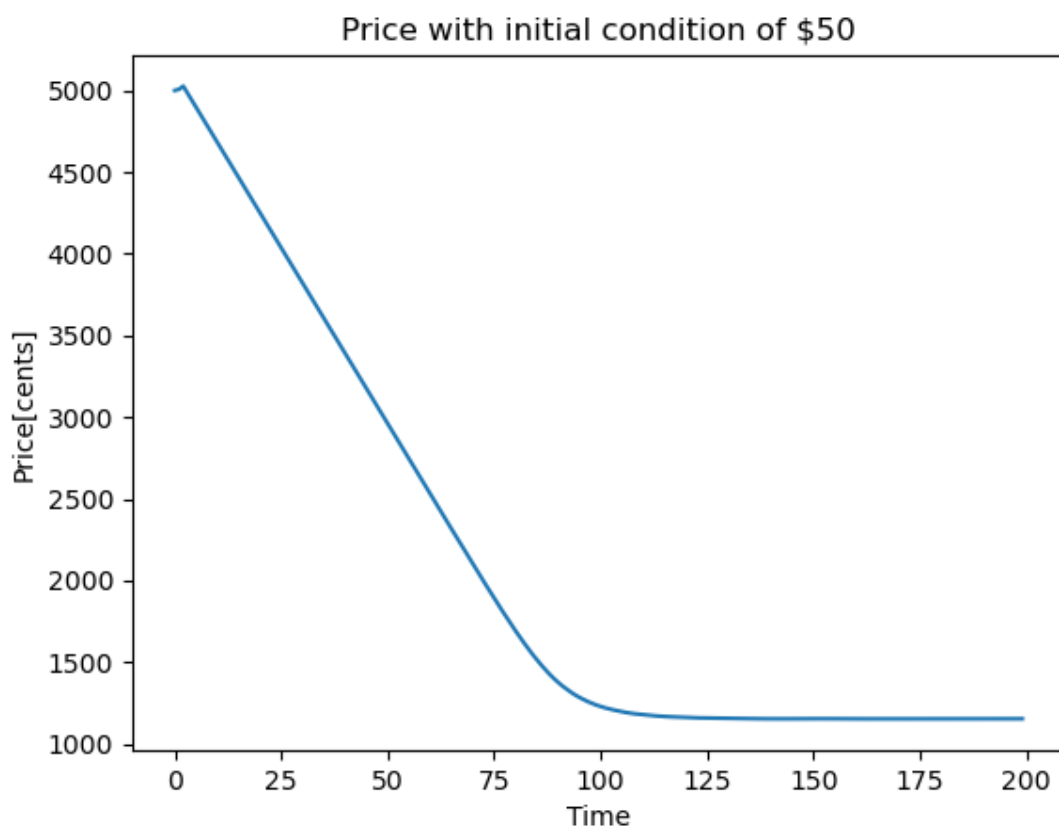


Figure 11: Convergence of price with initial price of \$50

Therefore, the amount offered is being changed from \$11.60 from task 3 to \$11.55.